Roll No. : .....

ODD END SEMESTER EXAMINATION JUNE-2025

Name of the Program: M. Tech **Semester: 2nd**

Course Name: Advanced Software Engineering **Course Code: MCS-201**

Time: 3 Hours **Maximum Marks: 100**

Note:

i. All questions are compulsory.
ii. Answer any two-question among a, b and c in each main question.
iii. Total marks for each main question are twenty.

Q1	**(20 Marks)**	
(a)	What is Software Requirement Specification (SRS), and why is it important in software development?	CO1
(b)	What are the main activities involved in software engineering, and what skills are needed to perform them?	CO1
(c)	What are the main challenges faced by software engineers in modern software development?	CO1
Q2	(20 Marks)	
(a)	What are the main principles of user-interface design?	CO2
(b)	How does good architectural design make software easier to maintain and scale?	CO2
(c)	What are some techniques used to ensure real-time performance in a distributed environment	CO4
Q3	(20 Marks)	
(a)	Compare System Testing and Component Testing in terms of scope, objective, and techniques. Which testing type is more suitable for detecting integration issues, and why?	CO6
(b)	What are the key steps involved in designing effective test cases? Explain how equivalence partitioning and boundary value analysis help in creating robust test scenarios.	CO3
(c)	Explain the three categories of software maintenance (Preventive, Corrective, and Perfective) with suitable examples. How does risk management play a role in planning and executing maintenance activities?	CO4
Q4	(20 Marks)	

(a)	Consider the following code snippet:
	void process(int x, int y) {
	  if (x > 0) {
	  if (y > 0) {
	  printf("Positive quadrant\n");
	  } else {
	  printf("Above X-axis\n");
	  }
	  } else {
	  if (y < 0) {
	  printf("Negative quadrant\n");
	  }
	  }
	  printf("Done\n");
	}
	Draw the Control Flow Graph (CFG) for the above code. Using the CFG, calculate the Cyclomatic Complexity.
		CO3

(b)	A software system has the following components: • External Inputs (EI): 6, with a complexity of average • External Outputs (EO): 4, with a complexity of high • Internal Logical Files (ILF): 3, with a complexity of average	CO3

	• External Interface Files (EIF): 2, with a complexity of low The following complexity weights are to be used: • EI (average): 4 • EO (high): 7 • ILF (average): 10 • EIF (low): 5 Calculate the total Function Point Count for this software system.	
(c)	How does the ISO 9000 standard influence software reliability and quality assurance practices in software development?	CO5
Q5	(20 Marks)	
(a)	How does software re-engineering help reduce the costs associated with long-term software maintenance and enhance the system's adaptability?	CO5
(b)	What are the major components of Software Quality Assurance (SQA) techniques that contribute to software reliability and quality?	CO2
(c)	What is the primary objective of Software Quality Assurance (SQA) in software development?	CO5